



From mice to humans

A multi-omic predictive framework for translational immunology

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September 2026

Mice are the workhorse — but how well do they *actually* predict human immunity?

- Murine models underpin preclinical vaccine research, yet their translational value remains **contested**, contributing to clinical trial failures.
- Prior cross-species comparisons were largely **descriptive**: gene-centric, linear, and disconnected from sequence evolution and cis-regulatory architecture.
- The consequence: **contradictory reports** and no practical framework to prioritise targets for human translation.

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MOUSE IMMUNIZATION
DATASETS SCREENED
FROM GEO (FIG. 1A)

6

MATCHED MOUSE–HUMAN
CONDITIONS CARRIED
FORWARD

One framework, three stimulus categories, matched across species

- **Vaccination** — Fludac® (influenza), Engerix B® (hepatitis B)
- **Infection** — *S. aureus*, *E. coli*
- **Injury** — burn, trauma-haemorrhage
- **+ DMD** (mdx mouse) as a biological negative control

Blood transcriptomes · GEO / BioProject
limma + empirical Bayes · covariates (age, sex, ethnicity)
human downsampled to n = 5, 200 iterations, median statistics

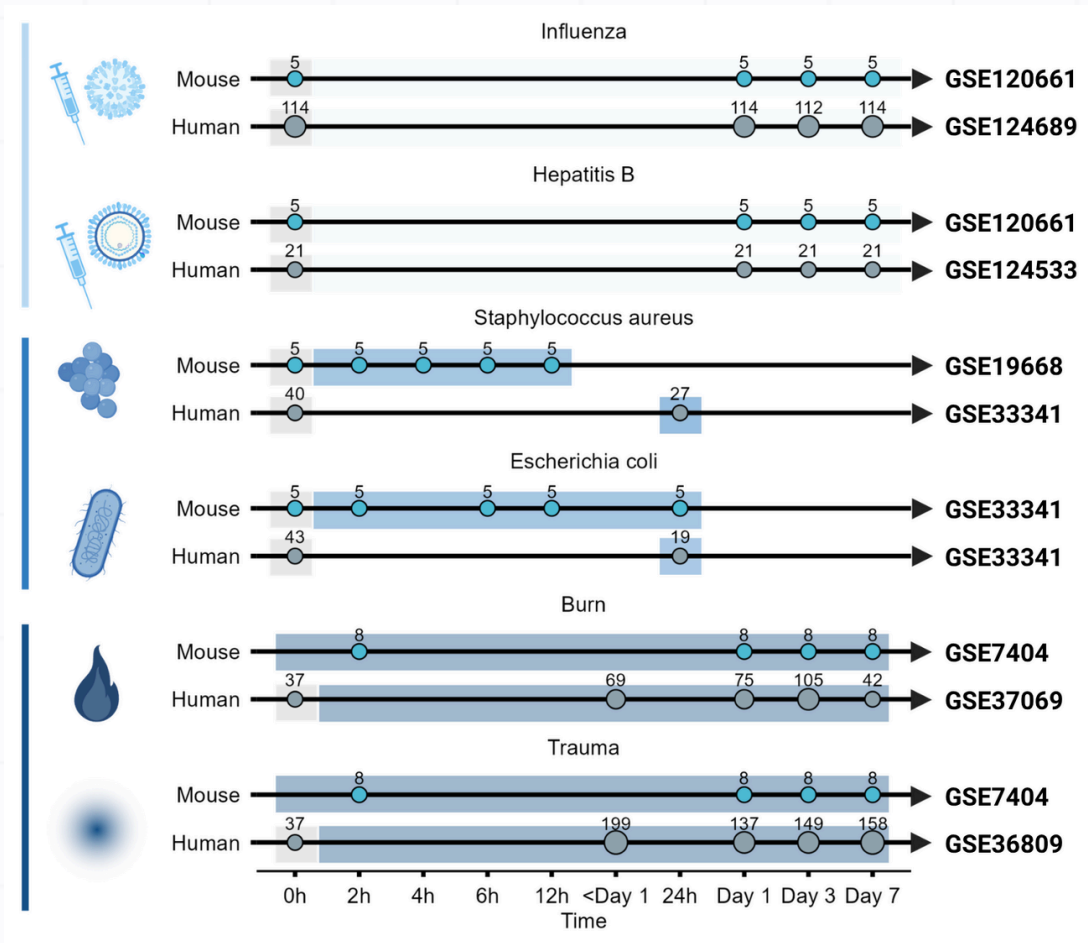


Fig. 1b — experimental timelines for matched mouse and human datasets across vaccination, infection and injury (GSE120661 · GSE124689 · GSE19668 · GSE33341 · GSE7404 · GSE37069 · GSE36809).

Cells disagree. Pathways agree.

0.89–0.97

MODULE-LEVEL (BTM
NES) PEARSON
CORRELATION
BETWEEN SPECIES

- Gene-level effect sizes correlate only **moderately**; module activation is **highly conserved**.
- Shared leading-edge genes = **67.5–79%** of mouse LEGs in enriched modules (70.5% pooled).
- Collapsing dimensionality into BTMs **reduces noise** and exposes a robust translational signature.



Fig. 4a–c — BTM module-level correlation (a) and classification performance (b, c) across conditions. · Fig. 5a — shared leading-edge genes across BTM process categories.

Translatability scales with the *strength* of the perturbation

Infection & injury — conserved

$\rho = 0.41\text{--}0.59$

Acute *S. aureus*, *E. coli* and burn engage evolutionarily stable, highly conserved programs.

Effect-size difference ($|\Delta\text{Log2FC}|$) significantly smaller than negative controls —
WMW adjusted $P < 1 \times 10^{-15}$.

Timing $\neq$ chronology · mouse 2–6 h matches the human peak inflammatory window (days 1–3) → align by biological equivalence

Single-dose vaccination — diverges

$\rho = -0.32\text{--}0.27$

Fluad® and Engerix B® fail to reach the systemic activation threshold, remaining near baseline controls.

Preclinical immunogenicity may require **boosters or stronger adjuvants** to emerge from species-specific noise.

Mouse signatures call
human direction — best at
module level

0.84–0.98

ROC-AUC · MOUSE
BTM SIGNATURES
CLASSIFYING HUMAN
MODULE REGULATION

0.64–0.74

ROC-AUC · GENE-
LEVEL MOUSE
SIGNATURES IN
INFECTION & INJURY

Biological negative control (DMD): ROC-AUC 0.48–0.61
— near random

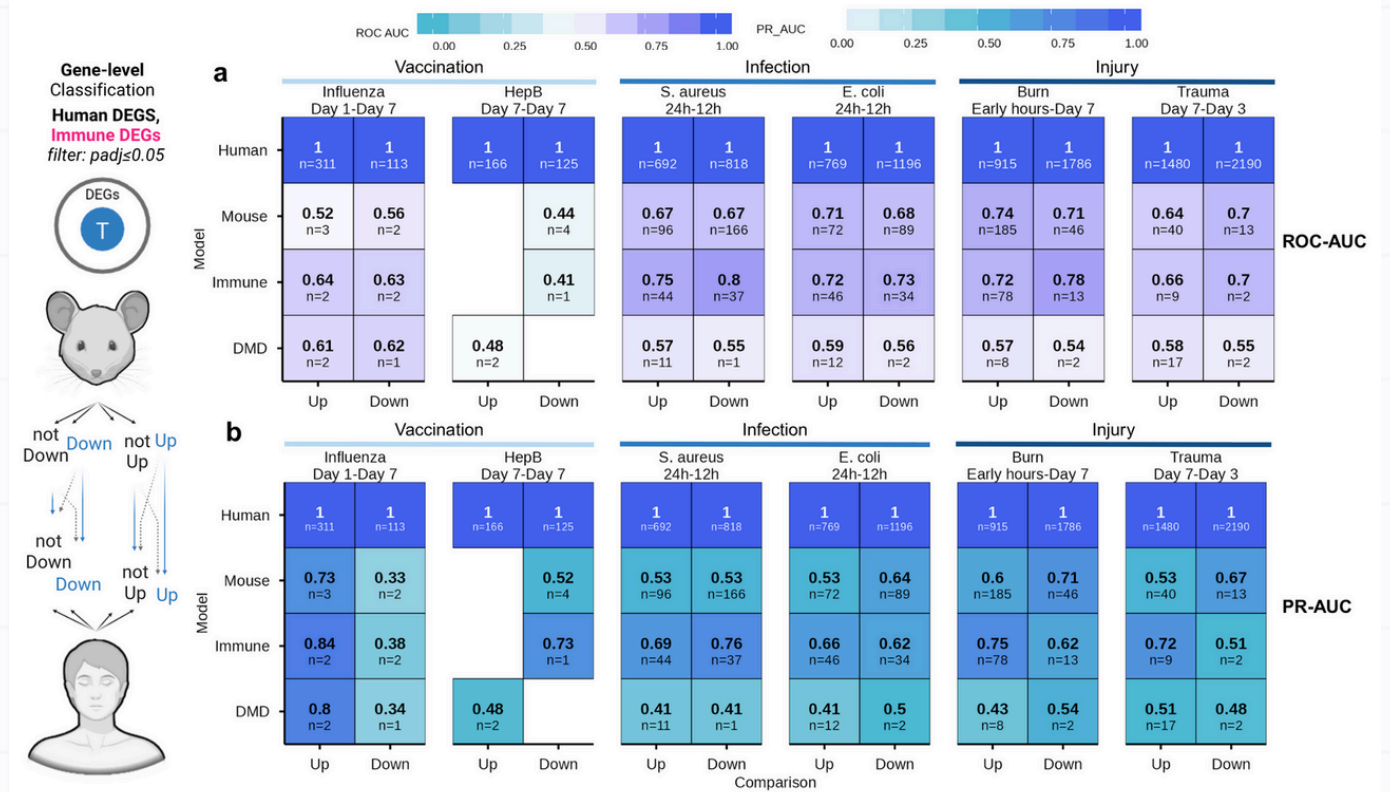


Fig. 3a–b — ROC-AUC and PR-AUC matrices for mouse signatures classifying up- and downregulated human genes, with human self-prediction and the DMD negative control.

Why genes *diverge*

Small per-gene effects, but a consistent direction: sequence constraint tracks **stability**; regulatory plasticity tracks **divergence**.

Associated with lower $|\Delta\text{Log2FC}|$

- Higher protein sequence identity $\rho = -0.08$
- Greater TF availability and TF sharing $\rho = -0.017 / -0.013$
- Conserved promoter-like (PLS) signatures n.s.

Associated with higher $|\Delta\text{Log2FC}|$

- Greater Kimura (K80) evolutionary distance $\rho = 0.10$
- Enhancer multiplicity — pELS / dELS counts $\rho = 0.036 / 0.039$
- CRE turnover and CTCF-bound pELS bound $\rho = 0.042$

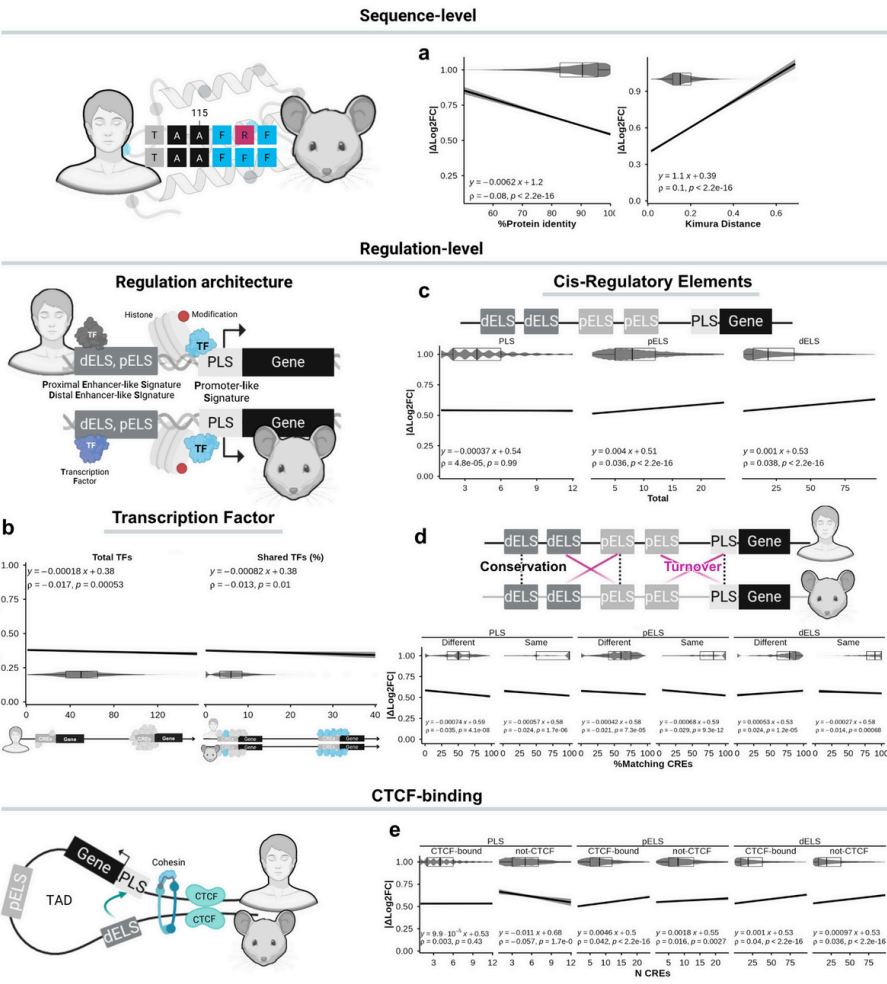


Fig. 6a–e — protein sequence identity and Kimura distance (a), transcription factor availability (b), cis-regulatory element abundance (c), CRE conservation and turnover (d) and CTCF binding (e) against $|\Delta\text{Log2FC}|$.

Layers turn near-chance transfer into *useful* prediction

- Mouse rank alone: $R^2 = 0.001$ · mouse direction alone: **ROC-AUC 0.511** — no transfer.
- evolution, TF, CRE and BTM layers → random forest $R^2 = 0.234$ for human rank and **ROC-AUC = 0.849** for direction.
- Shared vs mouse-only LEGs stay hard: lasso best by a whisker (**0.617** predictive / **0.627** explanatory).

LOCO-CV (4 folds) · nested layers: DGE Baseline → Full + BTM
4 algorithms: linear/logistic · lasso · random forest · neural network

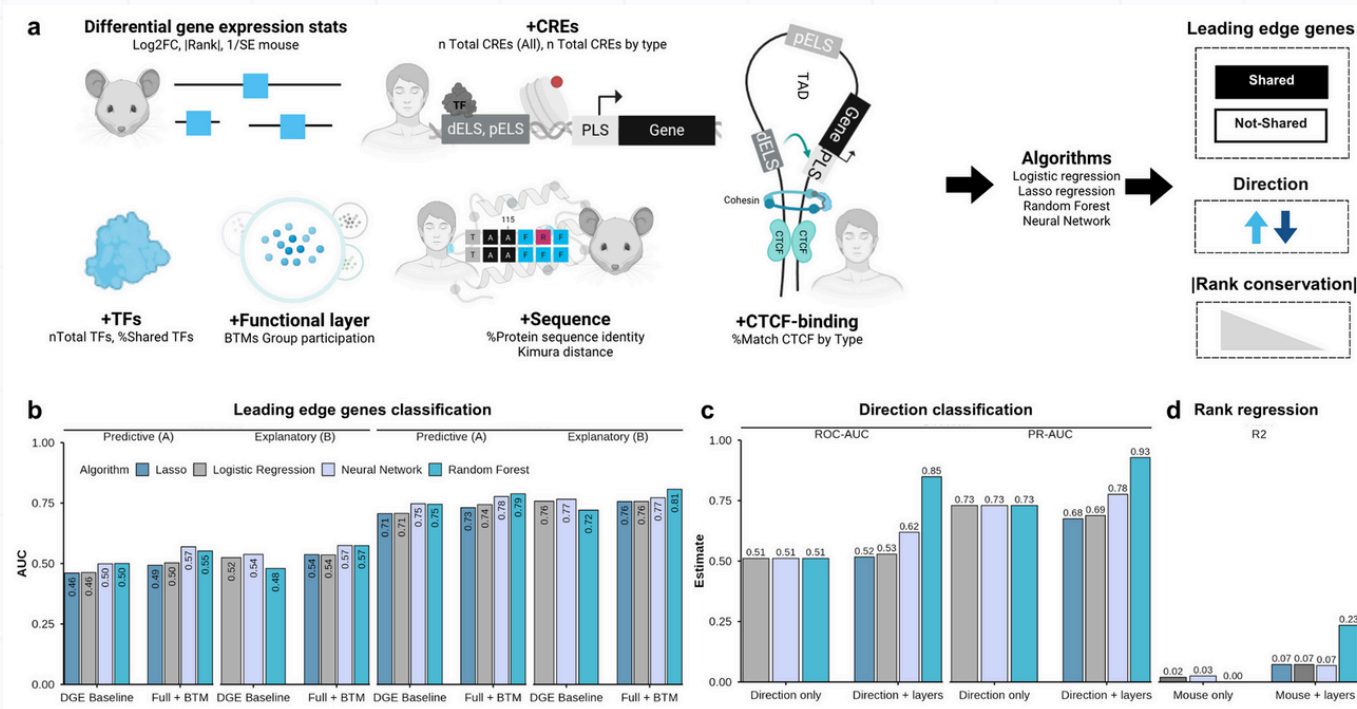


Fig. 7 — multilayer modelling: (a) layer integration workflow, (b) shared-LEG classification, (c) directional concordance and (d) human rank regression, across algorithms and feature layers.

Translatability is a function of *granularity*, *intensity* and *timing*

What the study establishes

- Module-level analysis beats gene-level analysis for cross-species concordance.
- Acute infection and injury translate; single-dose vaccination does not.
- Evolutionary and regulatory layers carry a conserved signal beyond mouse DGE.
- Four out-of-fold scores — `score_shared`, `score_rank`, `score_direction`, `score_translational` — prioritise candidate genes.

Limitations to keep in mind

- Small murine cohorts; scarce harmonised metadata.
- Two inbred strains only (C57BL/6, CB6F1) — strain choice shapes translatability.
- Single-dose mouse vaccination vs. human prime-boost regimens.
- Blood-based BTMs and one-to-one Ensembl orthologs — needs testing in other tissues and gene sets.

Open code, trained models, and a notebook to *apply* them

PIPELINE

mousetohuman_multilayer

Full R Markdown pipeline — curation, QC, limma DGE, GSEA, evolutionary and regulatory analyses, and the LOCO modelling script. Pinned with renv.

github.com/wapsyed/mousetohuman_multilayer

DOCUMENTATION

Project site

Overview, complete methodology — including the multilayer modelling pipeline — and the notebook map, in one place.

wapsyed.github.io/mousetohuman_multilayer

APPLY IT

mousetohuman_predict

Step-by-step notebook: from DGE input to GSEA, model prediction and visualisation of the four translational scores. Runs in RStudio via Binder.

github.com/wapsyed/mousetohuman_predict

Thank you — questions welcome.